DATA SCIENCE & ANALYTICS

MODULE 05 SESSION 19

DAX measures, filter context and CALCULATE

One measure, many answers: why the number changes.

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Mon 9 Nov 2026

BY THE END OF TODAY, YOU CAN

Class objectives: measures and their filter context

01 Write a reusable DAX formula

02 Divide safely with DIVIDE

03 Explain which filters affect a measure result

04 Add or replace a filter with CALCULATE

05 Compute a percent of total, checked against SQL

EVERYONE DO

Step 1.1

Warm-up quiz on D18’s star model

Five questions, live in Markdown Mash: grain, relationships, the date table, the dashed line, and one on COUNT(DISTINCT).

PART 1

Today’s model: D18’s star, rebuilt from files

Sunday’s tables as four files: you load them and draw the relationships again.



THE MODEL TODAY · PART 1
the star from four files

lime: loaded from the start files
dashed: inactive

INSTRUCTOR DO

Step 1.2

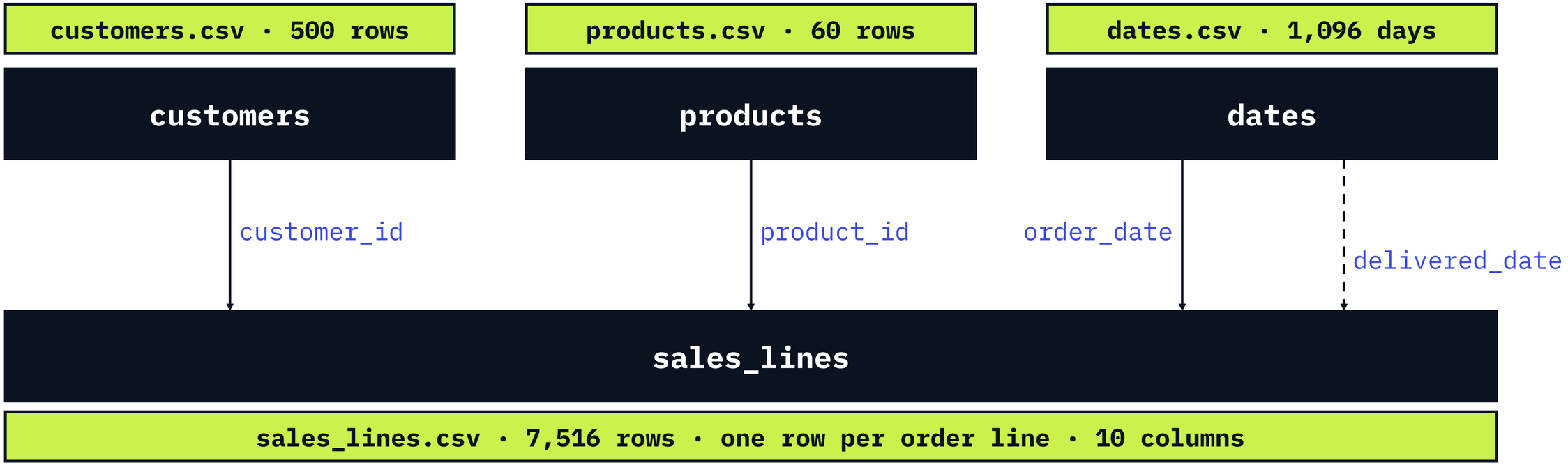
The Nakheel sales model starts from four CSV files

The same star as Sunday: sales_lines in the middle, customers, products and dates around it.

labs/D19/01-Evr-Load-And-Relate-Start-Files/README.md

TODAY’S MODEL · labs/D19/Resources/start · FOUR CSV FILES, D18’S FINISHED TABLES

Today’s model is D18’s star, loaded from four files



lime: the file you load · **solid line**: active · **dashed**: inactive (Sunday’s delivered date)

The four files are D18’s tables, already cleaned and merged. You load them and draw the four relationships again.

EVERYONE DO

Step 1.3

Activity 01: load the four start files and relate the tables

Load four files, draw four relationships, mark the dates table, and one card: Sum of line_value.

labs/D19/01-Evr-Load-And-Relate-Start-Files/README.md

PART 2

Measures, and how to write them in DAX

DAX is the formula language used to write measures in Power BI.



THE MODEL TODAY · PART 2
measures

lime: the measures you write on sales_lines

INSTRUCTOR DO

Step 2.1

Measures: what they are and how to write one

Implicit and explicit measures, columns and measures, then Total Revenue written one part at a time.

labs/D19/02-Evr-Base-Measures/README.md

FILTERS PANE

A filter can apply to a visual, a page or the report

Apply status = completed to the Overview card, the Overview page or all pages.

Filters on this visual

Overview page

Card	Table total
316,023.50	**340,370.00**

Orders page

Card
340,370.00

Filters on this page

Overview page

Card	Table total
316,023.50	**316,023.50**

Orders page

Card
340,370.00

Filters on all pages

Overview page

Card	Table total
316,023.50	**316,023.50**

Orders page

Card
316,023.50

Both cards and the table total sum line_value. Lime marks completed-only results.

Microsoft Learn: visual, page and report filters · checked 9 Nov 2026

DEFINITION · MEASURE

Measure: a named calculation run again for each cell

A measure is a named calculation, stored in the model, that Power BI runs again for every card, cell or bar that shows it.

ONE FORMULA, STORED ONCE

```
Total Revenue =
  SUM(sales_lines[line_value])
```

a card · Total Revenue

340,370.00

status	Total Revenue
completed	**316,023.50**
cancelled	**21,998.50**
pending	**2,348.00**
Total	**340,370.00**

The card adds every line. Each status row adds only lines with that status.

EXAMPLE · NAKHEEL · ORDER 100003 · THREE ORDER LINES

Implicit and explicit measures can sum the same rows

product_id	line_value
P-014	24.00
P-060	18.00
P-026	5.00

Sum of line_value

47.00

Total Revenue

47.00

The implicit sum comes from the visual. The explicit formula is reusable in DAX.

Capability	Implicit	Explicit
Sum or distinct count*	Yes	Yes
Named formula used in another measure	No	Yes
Custom DAX, such as a ratio	No	Yes

* Available implicit aggregations depend on the field type. Both sums here use the same three lines.

Microsoft Learn: Power BI aggregations · checked 8 Nov 2026

A column stores a value for each order line

For order 100003, the three line values add up to 47.00.

Product key	Quantity	Price	line_value
P-014	1	24.00	24.00
P-060	1	18.00	18.00
P-026	1	5.00	5.00

Total Revenue · order 100003 selected

47.00

Total Revenue · no filters

340,370.00

Add a custom column in Power Query before you load the data into the model.

A DAX calculated column evaluates a formula for each row in the model.

A report selection changes which lines the measure adds. The stored line values stay the same.

Microsoft Learn: calculation options · checked 9 Nov 2026

Total Revenue: the measure’s name comes first

The name is what the Data pane and every visual show. It may contain spaces.

```
Total Revenue =
    SUM(sales_lines[line_value])
```

Total Revenue: SUM adds up one column

SUM adds up every value of one column, on the rows the visual keeps.

```
Total Revenue =
    SUM(sales_lines[line_value])
```

Total Revenue: a column is written as table[column]

Write the table name followed by the column in square brackets.

```
Total Revenue =
    SUM(sales_lines[line_value])
```

in a card → 340,370.00

WHAT COMES BACK · Total Revenue IN A CARD · THE SAME NUMBER IN BIGQUERY

Total Revenue in a card reads 340,370.00

a card · Total Revenue

340,370.00

```
SELECT SUM(qty * price)
FROM nakheel.order_items;
```

→ 340,370.00, the same number in BigQuery

TOTAL REVENUE BY STATUS · THE THREE PARTS OF 340,370.00



cancelled 21,998.50 · pending 2,348.00

Every line of every status. D16’s 316,023.50 counts completed orders only: activity 02 gets there.

sales_lines HAS ONE ROW PER ORDER LINE (ITS GRAIN, D18) · ORDER 100003

Orders and Customers count distinct IDs, not rows

order_id	customer_id	product_id	line_value
100003	C-1357	P-014	24.00
100003	C-1357	P-060	18.00
100003	C-1357	P-026	5.00

sales_lines · order 100003's three rows

COUNTROWS(sales_lines) → **3 rows**

DISTINCTCOUNT(sales_lines[order_id]) → **1 order**

THE WHOLE TABLE · NO FILTER

COUNTROWS

7,516 rows

Orders · DISTINCTCOUNT(order_id)

3,000 orders

DISTINCTCOUNT(customer_id)

462 customers

One order, three lines. Count the order_id values, not the rows: SQL's COUNT(DISTINCT order_id).

EVERYONE DO

Step 2.2

Activity 02: write the base measures and check them against SQL

Select completed in the status slicer. Compare revenue, orders and customers with the SQL results.

labs/D19/02-Evr-Base-Measures/README.md

ACTIVITY 02, STEP 3 · THE SAME THREE MEASURES, BEFORE AND AFTER ONE SLICER

With status set to completed, the measures match SQL

NO SLICER

Total Revenue	Orders	Customers
340,370.00	3,000	462

slicer · status = completed The completed selection filters all three cards here.

Total Revenue	Orders	Customers
316,023.50 D16	2,771 D07 · completed orders	452 D09 · buying customers

A slicer filters the other visuals on its page by default. Here all three cards use completed sales lines.

INSTRUCTOR DO

Step 2.3

Average order value: divide revenue by orders with DIVIDE

Why AVERAGE gives the wrong answer on sales_lines, and how DIVIDE divides safely.

labs/D19/02-Evr-Base-Measures/README.md

A COMMON MISTAKE · AVERAGE ON A TABLE OF ORDER LINES

Average order value is revenue divided by orders

ORDER 100003 · THREE LINES

24.00	18.00	5.00

AVERAGE → 15.67
the average line

47.00 ÷ 1 order → 47.00
the average order

THE WHOLE TABLE · STATUS = COMPLETED

AVERAGE(sales_lines[line_value])	316,023.50 ÷ 6,951 lines	**45.46**
DIVIDE([Total Revenue], [Orders])	316,023.50 ÷ 2,771 orders	**114.05**

The grain of sales_lines decides what AVERAGE averages: lines. Divide the two measures to get the average order.

DIVIDE AND THE / OPERATOR · ROW 2: AN ILLUSTRATIVE EXAMPLE

DIVIDE returns a blank where the slash returns Infinity

The division	with /	with DIVIDE
316,023.50 ÷ 2,771	114.05	114.05
47.00 ÷ 0	**Infinity**	**(blank)**
(blank) ÷ (blank)	(blank)	(blank)

BLANK is an absent value. Its arithmetic and comparisons differ from SQL NULL.

The denominator is the number you divide by. By default, DIVIDE returns BLANK if it is zero or BLANK.

Microsoft Learn: DAX glossary and BLANK · checked 8 Nov 2026

Average Order Value: DIVIDE(top, bottom)

The inputs go inside parentheses. Give DIVIDE the amount first, then the denominator.

```
Average Order Value =
    DIVIDE([Total Revenue], [Orders])
```

A measure inside a measure goes in square brackets

Measures here use [Total Revenue] and [Orders]. Columns here use table[column]. This convention makes them easy to distinguish.

```
Average Order Value =
    DIVIDE([Total Revenue], [Orders])
```

no slicer → 113.46 · completed → 114.05

Microsoft Learn: column and measure references · checked 8 Nov 2026

WHAT COMES BACK · Average Order Value · CHECKED AGAINST D04 AND D08

Average Order Value reads 114.05 for completed orders

no slicer · 340,370.00 ÷ 3,000

113.46

status = completed · 316,023.50 ÷ 2,771

114.05

D04 KPI 3 · D08 average_order

```
SELECT AVG(order_total)
FROM nakheel.orders
WHERE status = 'completed';
```

→ 114.05, D04’s third KPI

SQL can use AVG(order_total) because nakheel.orders has one row per order.

EVERYONE DO

Step 2.4

Activity 02, step 4: Average Order Value with DIVIDE

One measure built from two measures. The completed slicer from step 3 is still on.

labs/D19/02-Evr-Base-Measures/README.md

THE RULE TO KEEP

A measure is written once, and Power BI runs it again for every card, cell and bar.

Total Revenue, Orders, Customers and Average Order Value: four measures, each checked against SQL.

PART 3

Filter context changes the rows used by a measure

Read order 100003 in the card and table, then select online in the channel slicer.



THE MODEL TODAY · PART 3
filter context

lime: the filters that reach sales_lines from each table

INSTRUCTOR DO

Step 3.1

Filter context in a card and a category table

Read the filters on one Outerwear cell, then follow them to the sales lines.

labs/D19/03-Evr-Filter-Context-And-Calculate/README.md

ONE REPORT BEFORE AND AFTER

The channel slicer changes the card and table

Page filter · status = completed

Channel slicer · no channel selected

Total Revenue · card

316,023.50

Category table · Outerwear row shown

Category	Total Revenue
Outerwear	**76,637.00**

Order 100003 contributes 47.00 to the card and 24.00 to Outerwear.

Page filter · status = completed

Channel slicer · online

Total Revenue · card

110,685.50

Category table · Outerwear row shown

Category	Total Revenue
Outerwear	**25,315.50**

Order 100003 is a store order. Its lines are excluded.

The completed filter stays in place. Selecting online limits both visuals to online sales.

FILTER CONTEXT

Filter context for the Outerwear revenue cell

Filter context means the filters in effect when Power BI calculates a value.

Filters on this page

`status = completed`

Channel slicer

`channel = online`

Category on this table row

`category = Outerwear`

The same category table · two rows shown

Category	Total Revenue
Outerwear	**25,315.50**
Bottoms	16,489.50

Power BI adds the line values for completed online Outerwear sales to get 25,315.50.

Microsoft Learn: filter context · checked 8 Nov 2026

SAME OUTERWEAR CELL AS PAGE 32

The Outerwear cell shows completed online revenue

`products[category] = Outerwear`

`status = completed`

`channel = online`

Example product rows

Key	Category
P-014	Outerwear
P-015	Outerwear
P-060	Sportswear

The relationship matches product_id.

Four example sales lines

Order	Key	Status	Channel	Amount	Included?
100003	P-014	completed	store	24.00	No (store)
100075	P-015	cancelled	online	60.00	No (cancelled)
100013	P-015	completed	online	**120.00**	**Yes**
100105	P-060	completed	online	54.00	No (category)

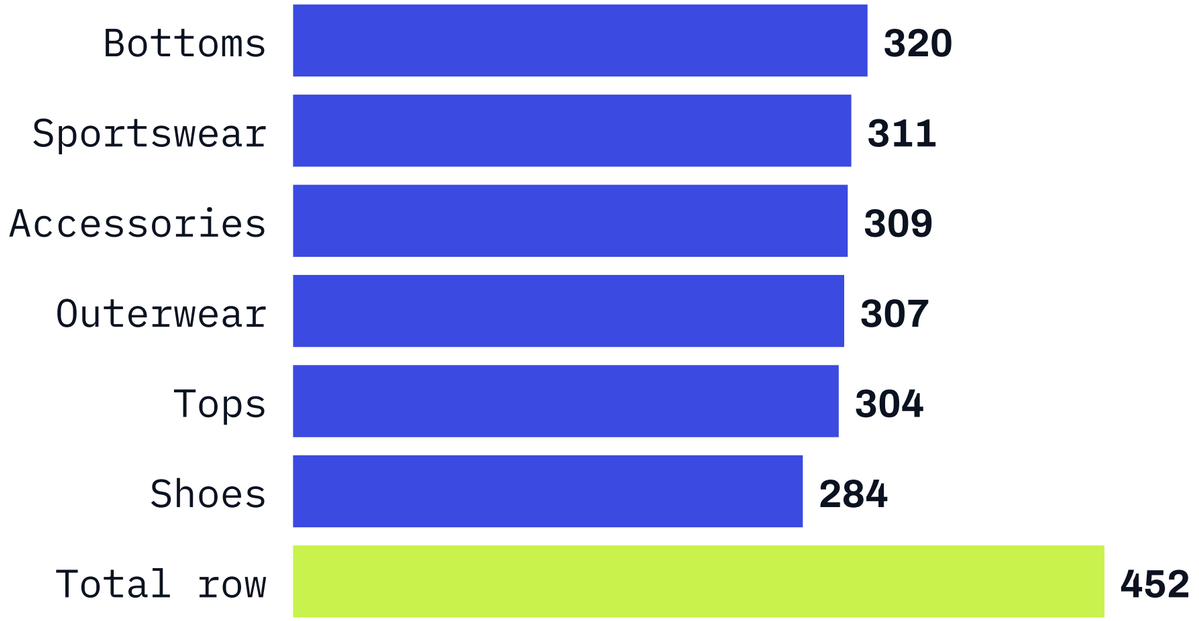
Whole dataset · SUM of 356 matching line values

25,315.50

Power BI matches product_id to find the Outerwear lines. It checks status and channel in sales_lines.

Customers = DISTINCTCOUNT(sales_lines[customer_id]) BY CATEGORY · STATUS = COMPLETED

The Customers total counts each buyer once



The rows add up to 1,835.

The total row counts each customer once: 452.

C-1357 bought in all six categories and counts once in each. The total counts that buyer once.

The Total cell has no category filter from a row. It counts each buyer once across the categories in scope.

QUICK CHECK

Quick check: which rows does the Shoes cell use under the online slicer?

slicer · channel = online

page filter · completed

category	Total Revenue
Outerwear	25,315.50
Shoes	?
Bottoms	16,489.50

Name the filters on the Shoes cell. Which rows of sales_lines are left for SUM?

QUICK CHECK · ANSWER

Completed online Shoes sales total 25,120.50

`category = Shoes`	from the table's row, through products	✓
`channel = online`	from the channel slicer, on sales_lines itself	✓
`status = completed`	from the page filter, on sales_lines itself	✓

left for SUM: 359 lines of 314 online orders → 25,120.50

channel and status are columns of sales_lines, so their filters need no relationship.

EVERYONE DO

Step 3.2

Activity 03: revenue and customers by category and channel

Use the completed status slicer for the lab. It supplies the same filter as the page filter in the example.

labs/D19/03-Evr-Filter-Context-And-Calculate/README.md

INSTRUCTOR DO

Step 3.3

CALCULATE: add a filter or replace one

Evaluate Total Revenue with completed status. Compare a table grouped by status with one grouped by category.

labs/D19/03-Evr-Filter-Context-And-Calculate/README.md

REPORT FILTER AND MEASURE RULE

Completed Revenue includes only completed orders

Same report page · no status filter or slicer selection

Total Revenue · all statuses

340,370.00

Completed Revenue · completed only

316,023.50

Use a page filter when both cards should show completed orders.

Use Completed Revenue when this calculation should always include only completed orders.

A visual filter set to completed would give the Total Revenue card 316,023.50.

CALCULATE evaluates a measure with changed filters

CALCULATE evaluates an expression with a changed filter context. A filter can be added or replace a filter on the same column.

```
CALCULATE(
    [Total Revenue],
    sales_lines[status] = "completed"
)
```

Completed Revenue

316,023.50

No status slicer is needed.

Total Revenue is 340,370.00 before the completed filter. The source table stays unchanged.

Microsoft Learn: CALCULATE · checked 8 Nov 2026

Completed Revenue: argument 1 is the measure to run

The first argument is the measure to run: Total Revenue, written once in Part 2.

```
Completed Revenue =
    CALCULATE(
        [Total Revenue],
        sales_lines[status] = "completed"
    )
```

Completed Revenue: argument 2 is a filter on status

The second argument is a filter. DAX text goes in double quotes.

```
Completed Revenue =
    CALCULATE(
        [Total Revenue],
        sales_lines[status] = "completed"
    )
```

with no slicer → 316,023.50

WHAT COMES BACK · TOTAL REVENUE AND COMPLETED REVENUE · NO SLICER

Completed Revenue keeps the category filter and replaces the status filter

BY CATEGORY · THE CATEGORY FILTER IS KEPT

category	Total Revenue	Completed
Outerwear	81,920.50	**76,637.00**
Shoes	67,710.00	**62,373.50**
Bottoms	55,099.50	**50,535.00**
Accessories	46,163.50	**43,525.00**
Sportswear	44,887.00	**42,365.00**
Tops	44,589.50	**40,588.00**

BY STATUS · THE STATUS FILTER IS REPLACED

status	Total Revenue	Completed
completed	316,023.50	**316,023.50**
cancelled	21,998.50	**316,023.50**
pending	2,348.00	**316,023.50**

Each row filters status to its own value. CALCULATE replaces that filter with status = completed.

Next, clear product filters to compare each category with completed revenue across all categories.

D18 · LAB 04 · APRIL 2026 · STATUS = COMPLETED

Delivered revenue uses the delivery date

```
Delivered revenue =
CALCULATE(
    SUM(sales_lines[line_value]),
    USERELATIONSHIP(sales_lines[delivered_date],
                    dates[date])
)
```

1 · the expression to evaluate

2 · the delivered-date path carries the April filter for this calculation



The same April selection gives 20,237.00 by order date and 17,815.50 by delivery date.

THE RULE TO KEEP

CALCULATE runs a measure with the filters you write: it adds a filter, or replaces the filter on the same column.

Completed Revenue: 316,023.50 with no slicer, on every status row.

EVERYONE DO

Step 3.4

Activity 03, step 4: Completed Revenue with CALCULATE

Clear the status slicer, write Completed Revenue, and compare it with Total Revenue by status.

labs/D19/03-Evr-Filter-Context-And-Calculate/README.md

REVIEW

Time's up.
Let's review.

Activities 02 and 03: the measures and their SQL

`Solved: labs/D19/03-Evr-Filter-Context-And-Calculate/Solved/solution.md`

Break

Then: percent of total

After the break: each category’s share of completed revenue, and Lab 04.

PART 4

Percent of total with CALCULATE and REMOVEFILTERS

Each category’s share of completed revenue, checked against D10’s window query.



THE MODEL TODAY · PART 4
percent of total

denominator
clears product filters

INSTRUCTOR DO

Step 4.1

Percent of total: clear product filters for the denominator

Compare completed Outerwear revenue with completed revenue across all categories.

labs/D19/04-Lab-Percent-Of-Total/README.md

A COMMON MISTAKE · DIVIDE([Completed Revenue], [Completed Revenue])

Dividing each category’s revenue by itself gives 100% here

category	Completed Revenue	divided by	share
Outerwear	76,637.00	76,637.00	**100%**
Shoes	62,373.50	62,373.50	**100%**
Bottoms	50,535.00	50,535.00	**100%**
Accessories	43,525.00	43,525.00	**100%**
Sportswear	42,365.00	42,365.00	**100%**
Tops	40,588.00	40,588.00	**100%**

Each shown category has nonzero revenue. Both amounts use the same category filter.

For this share, divide by completed revenue across all categories.

THE TOP AND THE BOTTOM OF THE SHARE · THE OUTERWEAR ROW · STATUS = COMPLETED

For the bottom number, remove the products filter

NUMERATOR · Completed Revenue

product_id	category
P-014	**Outerwear**
P-011	**Outerwear**
P-060	Sportswear
P-026	Tops

Example rows shown

`category = Outerwear`

product_id

product_id	line_value
P-014	**24.00**
P-060	18.00
P-026	5.00
P-011	**42.00**

All matching lines

76,637.00

DENOMINATOR · All Categories Revenue

product_id	category
P-014	**Outerwear**
P-011	**Outerwear**
P-060	**Sportswear**
P-026	**Tops**

Example rows shown

Category filter cleared

product_id

product_id	line_value
P-014	**24.00**
P-060	**18.00**
P-026	**5.00**
P-011	**42.00**

All matching lines

316,023.50

76,637.00 ÷ 316,023.50 = 24.3% · `D10: revenue / SUM(revenue) OVER ()`

Both sums use completed sales. The denominator clears the category filter and the relationship stays active.

REMOVEFILTERS(products)

All Categories Revenue keeps the channel filter

Both examples use the Outerwear row of the category table.

Category slicer · Outerwear only

Completed Revenue · Outerwear

76,637.00

All Categories Revenue

316,023.50

76,637.00 ÷ 316,023.50 ≈ 24.3%

All Categories Revenue clears the filter on products[category].

Channel slicer · online only

Completed Revenue · Outerwear

25,315.50

All Categories Revenue

110,685.50

25,315.50 ÷ 110,685.50 ≈ 22.9%

All Categories Revenue keeps the filter on sales_lines[channel].

All Categories Revenue also removes product filters set in the Filters pane.

Microsoft Learn: CALCULATE modifiers and ALL · checked 8 Nov 2026

All Categories Revenue: REMOVEFILTERS(products)

REMOVEFILTERS clears product filters for this calculation. The completed rule still applies.

```
All Categories Revenue =
    CALCULATE(
        [Completed Revenue],
        REMOVEFILTERS(products)
    )
```

316,023.50 across categories · all channels

% of Total divides category revenue by All Categories Revenue

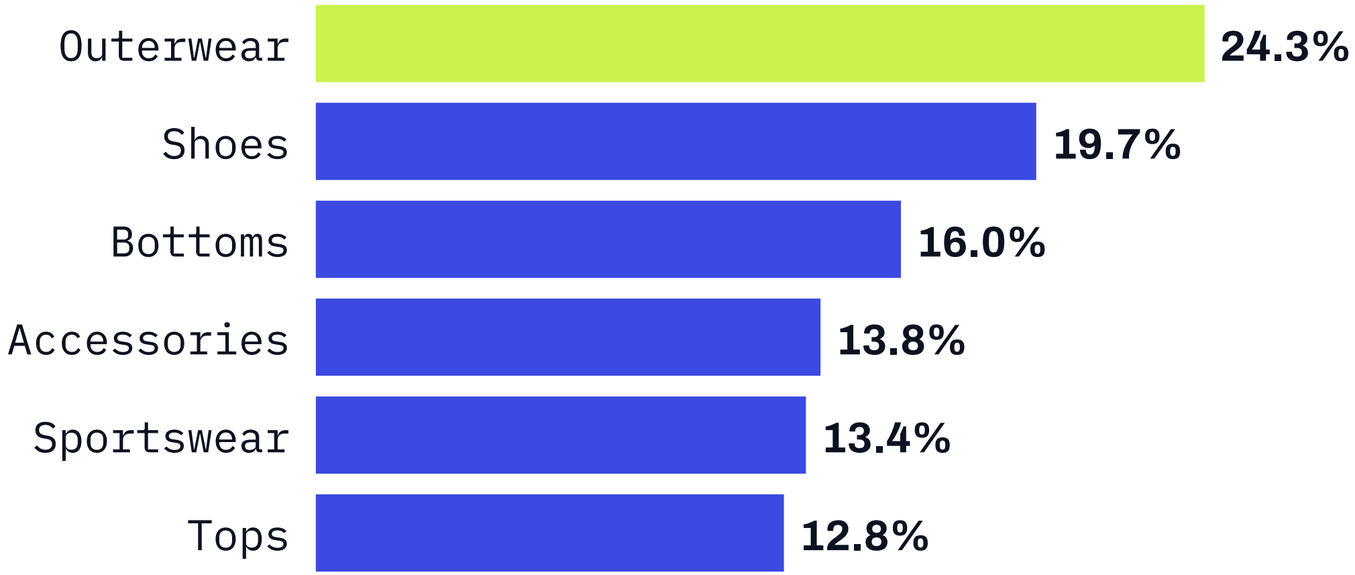
Two measures, so each can be checked. Format % of Total as a percentage.

```
% of Total =
    DIVIDE(
        [Completed Revenue],
        [All Categories Revenue]
    )
```

Outerwear → 24.3%

% OF TOTAL BY CATEGORY · COMPLETED REVENUE · OF 316,023.50

Each category’s share of completed revenue



total 100.0%

```
SQL check (D10):
revenue /
SUM(revenue) OVER ()
```

Outerwear brings almost a quarter of completed revenue. The six shares add up to 100%, as D10’s window query gave.

OUTERWEAR · SLICER channel = online · STATUS = COMPLETED · 25,315.50

Clearing channel in the denominator changes the share

REMOVEFILTERS(products[category])	÷ 110,685.50 = 22.9%	**completed online revenue**
REMOVEFILTERS(products)	÷ 110,685.50 = 22.9% (same)	**completed online revenue**
REMOVEFILTERS()	÷ 316,023.50 = 8.0%	**all completed revenue**

REMOVEFILTERS() also clears channel for the denominator. Completed Revenue still applies status = completed.

EVERYONE DO

Step 4.2

Lab 04: percent of total by category, checked against SQL

Build the share and compare the denominator under the category and channel slicers.

`labs/D19/04-Lab-Percent-Of-Total/README.md`

REVIEW

Time's up.
Let's review.

Lab 04: percent of total by category

`Solved: labs/D19/04-Lab-Percent-Of-Total/Solved/solution.md`

THE RULE TO KEEP

For each category’s share, divide its completed revenue by completed revenue across all categories.

With all six categories shown and no channel filter, the denominator is 316,023.50 on each row. The six shares total 100%.

Measures for revenue, counts and average order value

```
Total Revenue =
 SUM(sales_lines[line_value])
```

Adds the line values.

All statuses: 340,370.00

```
Orders =
 DISTINCTCOUNT(
  sales_lines[order_id])
```

Counts each order once.

All statuses: 3,000

```
Customers =
 DISTINCTCOUNT(
  sales_lines[customer_id])
```

Counts buyers in the current filters.

All statuses: 462

```
Average Order Value =
 DIVIDE([Total Revenue],
        [Orders])
```

Divides revenue by orders.

All: 113.46 · completed: 114.05

The completed slicer also changes Orders to 2,771 and Customers to 452. Practice: Activity 02.

Microsoft Learn: measures and calculated columns · checked 8 Nov 2026

The share uses a different filter scope for its denominator

```
Completed Revenue =
 CALCULATE(
  [Total Revenue],
  sales_lines[status]
   = "completed"
 )
```

Applies the completed status.

316,023.50, all categories

```
All Categories Revenue =
 CALCULATE(
  [Completed Revenue],
  REMOVEFILTERS(products)
 )
```

Clears product filters.

316,023.50 across categories

```
% of Total =
 DIVIDE(
  [Completed Revenue],
 [All Categories Revenue]
 )
```

Divides Completed Revenue by All Categories Revenue.

76,637.00 ÷ 316,023.50

Outerwear is 24.3% with no channel filter. Under online it is 22.9%. Practice: Lab 04; date-role formula: page 44.

Microsoft Learn: CALCULATE and denominator scope · checked 8 Nov 2026

KEEPFILTERS keeps values that satisfy both status filters

KEEPFILTERS keeps the existing status selection and applies the new status condition too.

Selected statuses	Written condition	Values satisfying both
completed	completed	**completed**
cancelled		

Total Revenue on the retained completed rows

316,023.50

Only completed satisfies both conditions. The measure uses those completed rows.

Microsoft Learn: KEEPFILTERS · checked 8 Nov 2026

Completed Kept returns BLANK for the cancelled selection

```
Completed Revenue =
 CALCULATE(
  [Total Revenue],
  sales_lines[status] = "completed"
 )
```

```
Completed Kept =
 CALCULATE(
  [Total Revenue],
  KEEPFILTERS(
   sales_lines[status]
    = "completed"
  )
 )
```

Selected status	Completed Revenue	Completed Kept
cancelled	**316,023.50**	BLANK
completed	**316,023.50**	**316,023.50**

No row has both statuses. SUM returns BLANK. Other filters still apply.

Microsoft Learn: KEEPFILTERS · checked 8 Nov 2026

ALL can supply product rows as well as clear filters

```
CALCULATE(
 [Completed Revenue],
 ALL(products)
)
```

Outerwear denominator: 316,023.50

```
CALCULATE(
 [Completed Revenue],
 REMOVEFILTERS(products)
)
```

Outerwear denominator: 316,023.50

product_id	category
P-014	Outerwear
P-060	Sportswear
P-026	Tops

3 of the 60 supplied products

```
COUNTROWS(ALL(products))
```

product rows
60

REMOVEFILTERS clears filters only; it supplies no table.

The denominator examples have no channel filter. D20 explains FILTER and its row-by-row evaluation.

Microsoft Learn: ALL and table results · checked 8 Nov 2026

Study check: predict the filters and the numbers

Case	Predict the result or explain the context
Status	Clear other filters and select only cancelled. Predict Completed Revenue and Completed Kept.
Denominator	Outerwear online: 25,315.50. Divide by 110,685.50 (online) and 316,023.50 (all channels).
Grain	Order 100003 has lines 24.00, 18.00 and 5.00. Find the average line value and average order value.
Context	DAX row context means evaluating an expression for one current row. A table groups by category. Does its Outerwear row supply filter context or DAX row context?

Predict first. Answers and explanations are on the next page.

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Study answers for filters, shares and order averages

Case	Answer and explanation
Status	316,023.50; BLANK. The first replaces cancelled with completed; the second keeps both conditions, leaving no matching rows.
Denominator	22.9% online; 8.0% all channels. The numerator is unchanged. Removing the channel filter changes the question.
Grain	Line average: 47.00 ÷ 3 = 15.67, rounded. Order average: 47.00 ÷ 1 = 47.00.
Context	Filter context. The category grouping filters the measure. A DAX calculated column uses one current sales line. Build one on D20 page 33.

Practice: Activities 02–03 and Lab 04. Read all seven study pages.

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STUDY LIST · IN CLASS AND IN THE LABS

Today’s DAX functions and where to practise each

In class	What it does	Activity
SUM	adds up one column on the rows a cell keeps	02
DISTINCTCOUNT	counts each order or customer once	02
COUNTROWS	counts rows: lines, not orders	02
AVERAGE	averages rows: the average line	02
DIVIDE	divides; a blank instead of Infinity	02
CALCULATE	runs a measure with filters you write	03
REMOVEFILTERS	clears a table’s filters inside CALCULATE	Lab 04
ALL	clears filters; can also return a table	Lab 04
USERELATIONSHIP	switches on an inactive relationship	D18

Practice	After class, self-guided, not submitted
04	**Percent of total: finish it if class ran out of time**
05	**Optional, for PL-300: each product’s share of its category**
Study	**Required: pages 61–67. KEEPFILTERS worked on 63–64; FILTER and row context: D20**

CLOSE

What you can do now with DAX measures

- Write a measure for filtered totals
- Use DIVIDE for ratios
- Identify a cell’s filters
- Change filters with CALCULATE

NEXT · WED 11 NOV Calculated columns, SUMX, time intelligence and report design.

Required study: pages 61–67.